

Scenario Set Questions 1

Scenario 1: A system checks if a user is eligible to vote based on their age.

Write logic to ask the user for their age and determine if they are eligible to vote based on whether they are 18 or older.

Solution:

Step 1: Ask the user to enter the age

Step 2: Verify the age using if conditional statement to check if the user is 18 or older.

Step 3: Check the “if output” and if “yes” -> print “Eligible to vote”

Step 4: If “no” then print “Not eligible to vote”

Scenario 2: A program processes a list of numbers and needs to find the largest value.

Write logic to identify and return the largest number from a given list.

Solution:

Step 1: Read the defined list containing the numbers

Step 2: Create an empty temp variable, assuming to be x, where the first number from the list is stored as the largest number.

Step 3: Using a for loop, iterate over the list values by comparing the current list value with the x

Step 4: If x is greater than the list value, keep the value in x and repeat the iteration

Step 5: If the list value is greater than x, then get that value and store it, update x with the list value

Step 6: Repeat the iterations until all the values in the list are compared, and finally print the x value, which is the largest number from the given list.

Scenario 3: A company provides employees with a 10% bonus if their salary exceeds \$50,000.

Write logic to determine the bonus amount based on the given salary.

Solution:

Step 1: Get the employee's salary and store it as "Salary"

Step 2: Verify the salary range using if conditional statement to check if the "salary" is greater than \$50,000 or not.

Step 3: Check the "if output" and if "yes" -> calculate the bonus of 10% of \$50,000 and print the "Bonus Based Salary"

Step 4: If "no" then set the bonus to be zero and print the "Salary"

Scenario4: A program evaluates a number to determine if it is even or odd.

Write logic to check whether a given number is even or odd.

Solution:

Step 1: Ask the user to enter the number

Step 2: Verify the number using if conditional statement to check if the %2 of number is equal zero or not.

Step 3: Check the "if output" and if "yes" -> print "given number is even"

Step 4: If "no" then print "given number is odd"

Scenario5: A text-processing tool reverses a given word or sentence for formatting purposes.

Write logic to take a word or sentence as input and produce its reversed version.

Solution:

Step 1: Ask the user to enter the word/sentence.

Step 2: Read the word/sentence and convert the string into a list of characters.

Step 3: Create an empty string as x, to store the reversed result, and Loop through each character in the original list

Step 4: Prepend each character to the result string from the list and print x

Scenario6: A grading system determines whether a student has passed or failed based on their score.

Write logic to check if a student has passed a subject by scoring at least 40 marks.

Solution:

Step 1: Ask the user to enter the student's mark.

Step 2: Verify the mark using if conditional statement to check if the mark is greater than 40% or not.

Step 3: Check the "if output" and if "yes" -> print "student is passed "

Step 4: If "no" then print "student is failed "

Scenario7: A retail store offers a 20% discount if a customer's total order exceeds \$100. Write logic to calculate the final amount to be paid after applying the discount.

Solution:

Step 1: Get the total order value and store it as "TOV"

Step 2: Verify the TOV range using if conditional statement to check if the "TOV" is greater than \$100 or not.

Step 3: Check the "if output" and if "yes" -> calculate the final amount with TOV - 20% of \$100 and print the "Discounted Amount"

Step 4: If "no" then set the discount to be zero and print the "TOV"

Scenario8: A banking system processes withdrawal requests and ensures the user has enough balance.

Write logic to check if a user has enough balance before allowing a withdrawal and update the remaining balance accordingly.

Solution:

Step 1: Get the withdrawal request from the customer and store the value as x

Step 2: Read the current account balance as y

Step 3: Verify the WithdrawalAction range using if conditional statement to check if the x is less than or equal to y.

Step 4: Check the “if output” and if “yes” -> process the withdrawal action and update the current balance after withdrawal as Y

Step 5: If “no” then dismiss the transactions printing “ No sufficient Balance int the account to make the withdrawal”

Scenario: A calendar system verifies whether a given year is a leap year based on standard leap year rules.

Write logic to determine whether a given year is a leap year.

Solution:

Step 1: Read the year as x

Step 2: Verify the leap year using if conditional statement to check if the x is divisible by 4. Check the “if output” and if “No” -> Not a leap year & “if output” and if “Yes” -> Go to step 3

Step 3: Verify the leap year using if conditional statement to check if the x is divisible by 100. Check the “if output” and if “No” -> It is a leap year & “if output” and if “Yes” -> Go to step4.

Step 4: Verify the leap year using if conditional statement to check if the x is divisible by 400. Check the “if output” and if “Yes” -> It is a leap year & “if output” and if “No” -> Not a leap year.

Step5: If otherwise, print “not a leap year”

*Scenario10: A program filters out only even numbers from a given list.
Write logic to extract and return only the even numbers from a list.*

Solution:

Step 1: Read the list.

Step2: Create an empty list as x

Step 3: Verify the list using if conditional statement and for loop to check if the numbers is divisible by 2 or not.

Step 4: Check the “if output” and if “yes” -> append the x with iterated current value and repeat step 3 with next value in the list

Step 5: If “no”, discard the iterated current value from the list and repeat step 3 with next value in the list

Step 5: Return the x which contains even numbers
